

CAPSTONE PROJECT

Air Quality Indicator Analysis

PRESENTED BY

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OUTLINE:

- **Problem Statement**
- **Proposed System/Solution**
- **System Development Approach**
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

Governments often react to pollution only when it happens (like a sudden winter smog), rather than preparing in advance and Most reports only show the past (what happened in 1995). They don't help leaders decide what to do in 2026. Decision makers lack a clear way to predict how specific policy changes will impact air quality, and traditional reporting hides the frequency of toxic health risks and data gaps across different regions.

PROPOSED SOLUTION:

- **Data Collection:**
 - Imported a multi-year air quality dataset containing historical readings of SO₂ and NO₂ across various Indian States
 - Identified that the raw data contained a mix of Industrial, Residential, and Sensitive area types, spanning from 1990s to the 2010s
- **Data Preprocessing:**
 - Clean and preprocess the collected data to handle missing or null values.
 - Feature engineering , created Is_Dangerous flag to categorize toxic vs. Safe days.
- **Deployment:**
 - Develop a user-friendly interface or Heatmap, Hotspot Table and Simulation Slider into a single user-facing interface.
 - Enabled interactions so that selecting a state instantly update graphs and cards.
- **Evaluation:**
 - Compared Simulated SO₂ against Actual SO₂ using Line Charts.
 - Evaluated the “Trust Score” of the data using the Treemap to ensure the conclusions weren’t based on “Blind Spots”.

SYSTEM APPROACH:

System Requirements

To ensure the dashboard runs smoothly with large environmental datasets, the following specifications are recommended:

RAM: Minimum 8GB

Processor: Quad-core i5

User Access: Power BI Service (Pro or Premium) if the dashboard is to be deployed to government stakeholders for cloud access.

The Modeling Library

- **DAX(Data Analysis Expressions):** This is the "Brain" of your system. It is responsible for the **Machine Learning Simulation** and **Insights**.
- **Power BI Visualization Library**
- **Power Query**

ALGORITHM & DEPLOYMENT:

The Classification Algorithm (Threshold Logic)

This algorithm categorizes air quality status for every row in the dataset.

- **Logic:** A Boolean-based classifier.
- **Formula:** $\text{Is_Dangerous} = \text{IF}(\text{SO}_2 > 50 \text{ OR } \text{NO}_2 > 40, 1, 0)$.
- **Purpose:** To transform continuous numerical data into a discrete "Health Risk" metric.

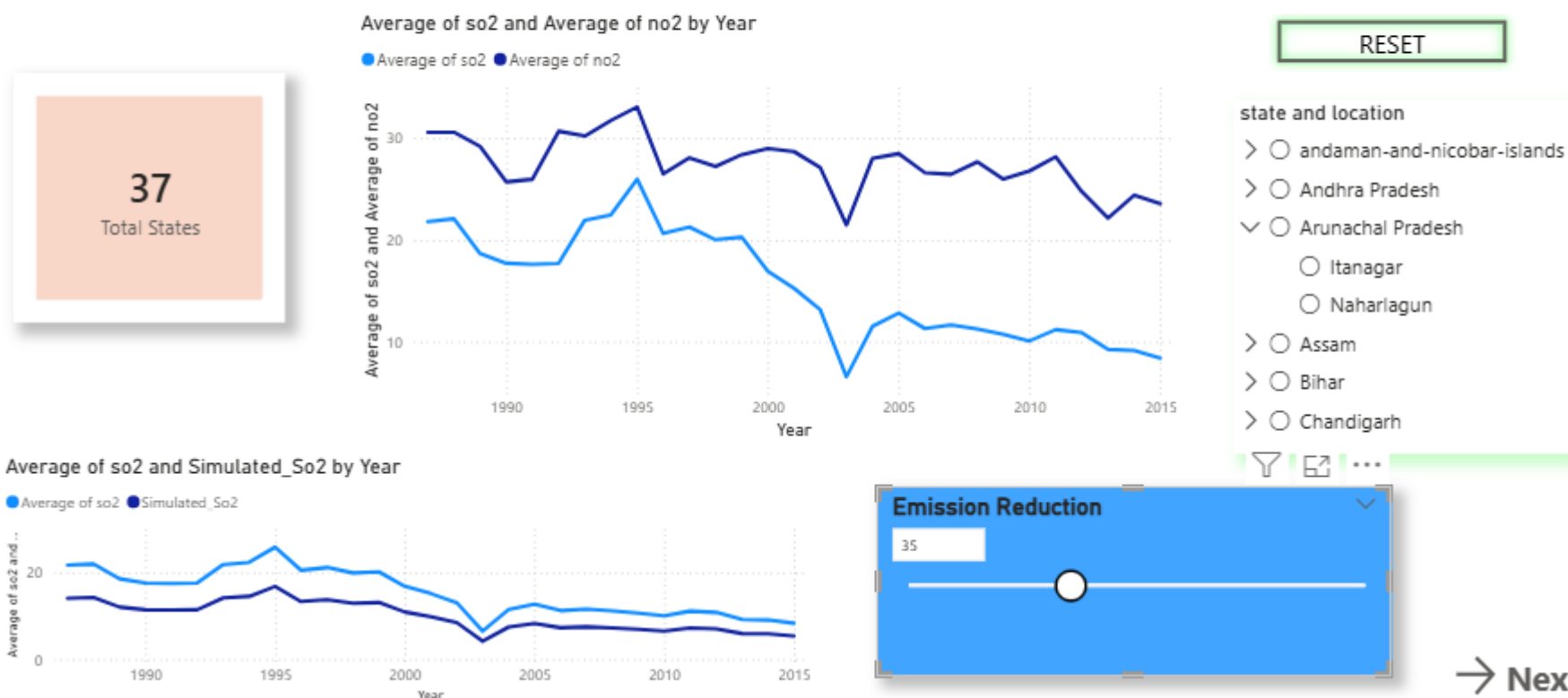
Dashboard Packaging

Individual visuals (the Heatmap, the Slider, the "Danger Days" chart) are "pinned" to a high-level Dashboard.

A **Mobile Layout** is configured so that environmental officers can check city "Hotspots" on their smartphones while in the field

RESULT:

Air Quality Indicator Analysis

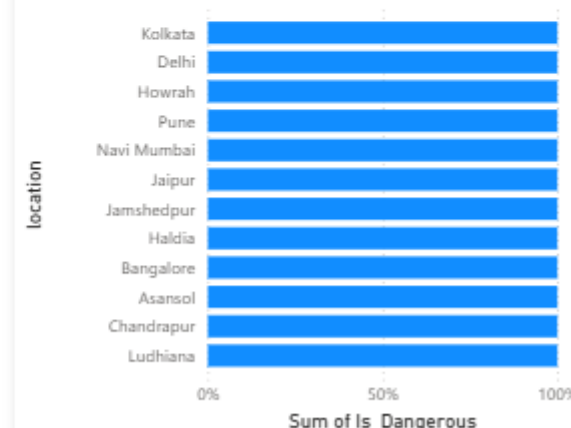


RESULT:

Year	January	February	March	April	May	June	July	August	September	October	November	December	Total	
	9.50												9.50	
1987	20.39	23.59	21.48	21.59	23.44	20.90	14.78	18.32	28.80	25.45	17.28	36.73	21.83	
1988	23.08	22.11	19.65	21.34	22.84	16.04	26.42	26.46	15.23	10.20	33.88	19.20	22.10	
1989	23.24	21.10	21.93	17.40	16.47	15.38	15.09	14.06	15.88	11.10		22.41	18.67	
1990	17.92	20.33	17.55	17.02	17.26	15.93	14.08	16.04	17.08	18.33	20.24	21.59	17.72	
1991	22.42	21.37	17.61	17.43	16.44	17.06	15.63	13.22	14.75	16.28	17.74	19.40	17.61	
1992	19.86	16.02	20.43	18.48	19.03	15.05	17.12	16.52	13.89	15.98	18.30	18.76	17.72	
1993	23.48	21.53	20.04	19.95	22.56	22.25	19.85	19.26	21.14	21.46	24.48	26.69	21.96	
1994	23.52	19.07	23.81	19.00	20.30	19.82	18.67	21.86	22.37		25.53	34.04	22.48	
1995	30.62	28.71	25.97	26.43	27.38	24.89	22.26	23.73	22.03	26.46	25.90	27.85	26.00	
1996	24.09	22.23	21.09	19.28	19.98	19.57	16.87	18.22	19.83	21.53	21.75	23.21	20.65	
1997	26.84	25.44	22.95	20.57	19.56	18.93	17.95	18.03	19.35	21.82	20.77	21.70	21.28	
1998	21.72	20.42	21.00	21.07	18.89	19.98	21.03	18.68	17.27	16.74	21.50	21.80	20.03	
1999	22.86	24.92	24.21	20.30	20.21	18.78	17.30	16.95	17.46	17.50	19.26	21.32	20.29	
2000	19.37	17.42	17.97	16.60	16.79	15.45	15.08	15.70	16.39	17.20	17.25	17.59	16.91	
2001	18.15	18.30	17.81	15.94	14.79	13.29	13.06	12.90	14.15	13.96	15.25	15.32	15.25	
2002	14.20	13.71	13.76	13.76	13.07	12.58	12.99	13.11	12.41	12.64	13.16	12.70	13.15	
2003	5.14	5.08	6.49	6.35	5.86	5.87	5.43	6.73	6.23	9.12	9.37	9.05	6.56	
2004	11.86	12.06	11.79	11.25	11.28	11.58	11.02	10.71	11.04	11.52	11.88	12.51	11.53	
2005	13.28	13.27	12.80	13.29	13.26	12.38	12.26	12.68	12.21	12.64	13.06	13.03	12.83	
2006	11.46	11.62	11.15	11.36	11.34	10.89	11.01	10.75	10.96	11.73	11.54	11.88	11.31	
2007	12.39	12.24	12.10	11.86	11.85	11.32	10.83	10.69	11.24	11.34	11.83	12.20	11.65	
2008	11.53	11.81	11.72	11.36	11.58	11.05	10.70	10.55	10.76	11.28	11.36	11.36	11.27	
2009	11.42	11.37	11.39	10.89	10.30	10.36	10.05	9.87	10.24	11.15	10.67	11.00	10.73	
2010	10.92	10.71	10.56	10.12	10.21	9.88	9.39	9.45	9.73	9.84	10.07	10.35	10.09	
2011	11.46	12.02	11.56	11.30	11.81	10.05	10.16	9.97	10.66	11.58	11.96	11.88	11.20	
2012	11.27	10.63	10.73	10.79	10.48	10.74	10.03	9.98	10.20	11.00	12.38	12.50	10.90	
2013	9.75	9.63	9.57	10.01	9.67	8.39	8.83	8.55	9.00	9.37	9.14	9.13	9.25	
2014	10.15	9.16	9.39	9.56	9.46	8.83	9.06	8.24	8.50	9.34	8.69	9.16	9.14	
2015	9.26	9.15	8.93	8.32	8.38	7.89	7.42	7.59	7.59	8.32	8.82	8.99	8.41	
Total	9.50	11.60	11.42	11.16	10.97	10.91	10.29	10.07	9.95	10.27	10.80	11.10	11.39	10.83

location	state	Average of so2
Ahmedabad	Gujarat	14.18
Bangalore	Karnataka	13.67
Chandrapur	Maharashtra	20.49
Chennai	Tamil Nadu	11.70
Jamshedpur	Bihar	47.30
Jamshedpur	Jharkhand	37.39
Kolkata	West Bengal	10.31
Nashik	Maharashtra	27.07
Navi Mumbai	Maharashtra	17.27
Pune	Maharashtra	24.68
Visakhapatnam	Andhra Pradesh	10.80
Total		16.70

Sum of Is_Dangerous by location

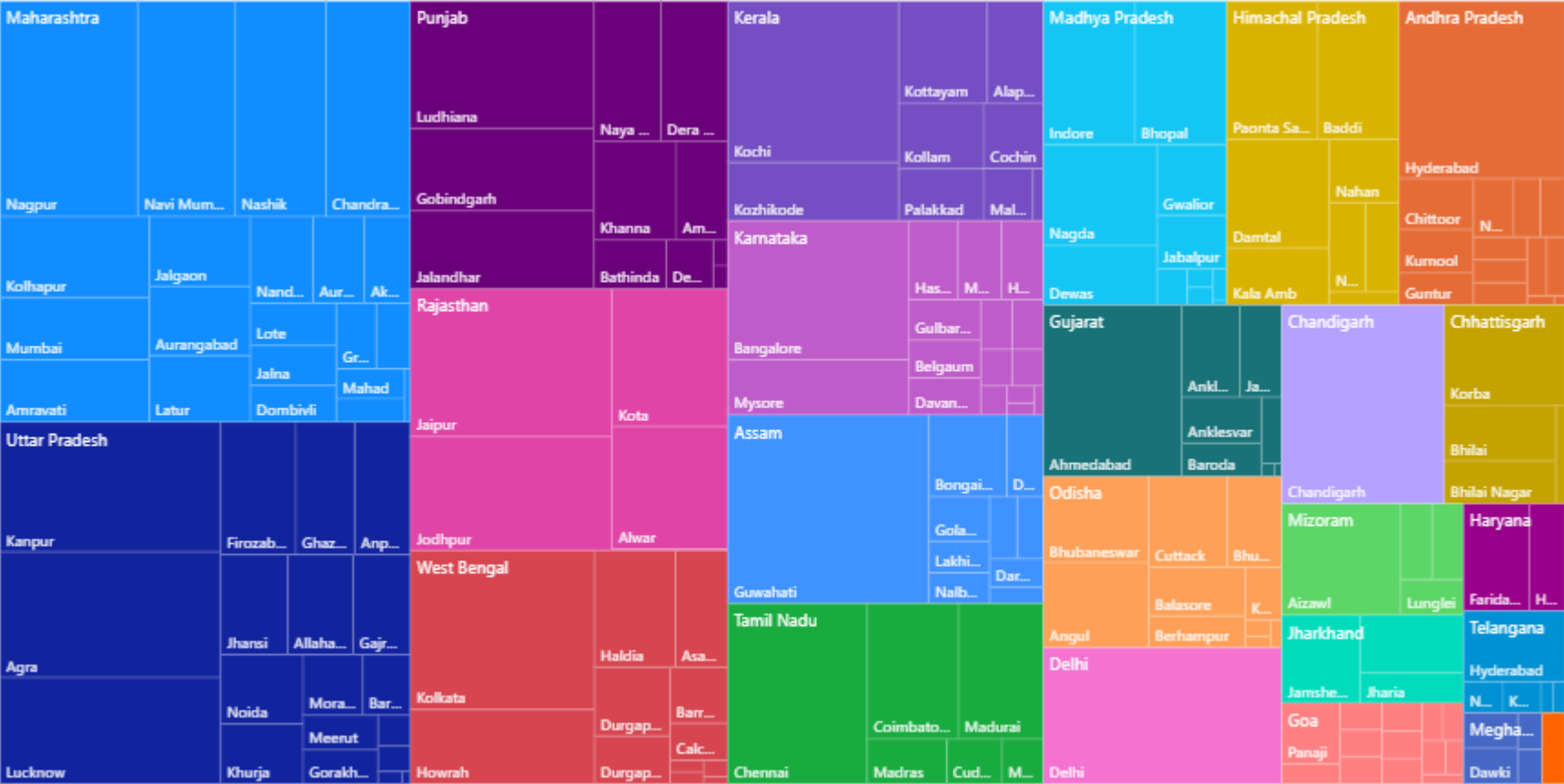


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RESULT:

Sum of Data Reliability by state and location



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CONCLUSION:

This project successfully transformed a vast, complex dataset of historical air quality records into a **Dynamic Decision Support System**. By shifting the focus from static historical reporting to interactive predictive modeling, we have provided a blueprint for data-driven environmental policy.

FUTURE SCOPE:

- The architecture we built for one city is a "Template." The future scope includes:
- **Geospatial Expansion:** Scaling the model to cover the entire Indo-Gangetic Plain or a national-level dashboard.
- **Automated Reporting:** Implementing **Power Automate** to send mobile alerts to city officials whenever a "Danger Day" threshold is crossed in real-time.

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REFERENCES:

GitHub Link: [Link](#)

Dataset used Link: [Link](#)

Thank You